Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	19 October 2026
Team ID	Roll 58
Project Name	EV Market Analysis Dashboard using Tableau Public
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect to the supplied EV Excel dataset and access fields for manufacturer, model, category, annual sales, battery type, battery capacity, range, price, manufacturing country, customer rating, and model year where available.	High
2	Data Processing	Validate data types, identify missing values and duplicates, and ensure sales, range, price, and rating fields are numeric. Use appropriate aggregation and exclude null values from relevant calculations.	High
3	Business Analysis	Calculate total recorded annual sales, distinct vehicle/model counts, average driving range, average customer rating, sales by manufacturer/category, and comparisons of price versus range.	High
4	Business Insights & Analytical Outputs	Provide eight visuals: sales by manufacturer; sales by category; battery technology distribution; average range by category; price versus range; customer ratings by manufacturer; manufacturing country analysis; sales by model year.	High
5	User Interaction & Analysis	Allow users to filter by manufacturer, vehicle category, battery type, manufacturing country, and model year where applicable. Enable chart selection to filter related views.	Medium
6	Data Update & Reporting	Refresh the workbook when an updated source file is available, maintain clear chart titles and units, and publish the finished	Medium

		dashboard on Tableau Public with a shareable link.	

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard must use clear titles, readable labels, consistent colors, and intuitive filters so non-technical business users can understand the insights.
NFR-2	Security	Use only non-confidential data. Tableau Public workbooks and published data are public and should not contain private or sensitive information.
NFR-3	Reliability	Calculations must use correct aggregations; missing values and duplicate records should be checked and documented.
NFR-4	Performance	Use aggregated charts and filters rather than displaying all 200,000 rows as individual marks; limit overly detailed views.
NFR-5	Availability	Once published successfully, the dashboard should be accessible through its Tableau Public URL when the service is available.
NFR-6	Scalability	Use reusable worksheets and consistent calculated fields so additional EV records or future columns can be incorporated with limited redesign.